Off-by-One Errors in Sequence Indexing
Answer Key

Algebra 1

Quick Answers

1. (a) the seventh term = 23, —
2. the step = 5
3. (a) the first term = 5, (b) the fifth term = 21, —
4. (a) how many terms = 15, —
5. (a) the eighth term = −9, (b) the term number = 11
6. (a) the correct sum = 78, (b) the student's sum = 77, —
7. (a) seats in row 20 = 52, (b) the row with 46 seats = 17
8. (a) the hundredth term = 695, (b) how many terms = 10, —

What is verified

3 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 3, 4, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSF-BF.A.2 – *problems 1, 2, 3, 4, 5, 6, 8*
HSF-LE.A.2 – *problem 7*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

1. *If they got 26: counted seven jumps instead of six*
2. *If they got 6.25: counted four jumps from term four to term nine instead of five*
4. *If they got 14: subtracted the two index numbers without adding one*
7. *If they got 54: used the row number as the number of steps*
8. *If they got 702: added the step one hundred times instead of ninety-nine*

1. $a_1 = 5$, $d = 3$; a student wrote $a_7 = 5 + 7(3) = 26$. (a) Correct a_7. (b) Name the error.

Solution (a): From term 1 to term 7 there are six jumps, not seven: $5 + 6(3) = 5 + 18$.

$$\boxed{a_7 = 23}$$

(b) — instructor-judged. A correct answer says reaching term seven from term one takes six jumps, so the step was added one time too many. Full credit for naming the jump count as one less than the term number; partial credit for saying only that the answer is three too big.

---

2. $a_4 = 20$, $a_9 = 45$; a student divided by 4. Find the step correctly.

Solution: Terms 4 and 9 are five jumps apart — count them: $4 \to 5 \to 6 \to 7 \to 8 \to 9$. So $d = \frac{45-20}{5} = \frac{25}{5}$. Check: $20 + 5(5) = 45$ ✓ $\boxed{d = 5}$

---

3. $a_n = 4n + 1$ for $n \geq 1$; a student read the first term off at $n = 0$. (a) a_1. (b) a_5. (c) Name the error.

Solution (a): The rule starts at $n = 1$, so substitute 1: $4(1) + 1$. $\boxed{a_1 = 5}$

Solution (b): $4(5) + 1 = 20 + 1$. $\boxed{a_5 = 21}$

(c) — instructor-judged. A correct answer says the rule is stated for term numbers starting at one, so substituting zero names a term that does not exist; the first term is what the rule gives at input one. Full credit for tying the mistake to the stated starting index; no credit for an answer that only reports the right number.

---

4. How many terms in $a_6, a_7, \ldots, a_{20}$? A student wrote $20 - 6 = 14$. (a) Correct count. (b) Why subtraction alone fails.

Solution (a): Subtracting the indices counts the jumps between listed terms; the list holds one more item than that: $20 - 6 + 1 = 14 + 1$. $\boxed{\text{count} = 15}$

(b) — instructor-judged. A correct answer explains that subtracting the two index numbers counts the gaps between listed terms, and a list has one more term than it has gaps, so one must be added back. A short illustration (a_6, a_7, a_8 is three terms but two gaps) earns full credit; stating the add-one rule with no reason earns partial credit.

---

5. $a_1 = 12$, $d = -3$; the student's rule was $a_n = 12 - 3n$. (a) Correct a_8. (b) Which term equals -18?

Solution (a): The correct rule is $a_n = 12 - 3(n-1)$, which gives 12 at $n = 1$ as it must. At $n = 8$ there are seven jumps: $12 - 7(3) = 12 - 21$. $\boxed{a_8 = -9}$

Solution (b): Solve $12 - 3(n-1) = -18$: then $3(n-1) = 30$, so $n - 1 = 10$. Check: $12 - 3(10) = -18$ ✓ $\boxed{n = 11}$

---

6. $a_n = 3 + 4(n-1)$; the student summed from $n = 0$ to 6. (a) Sum of the first six terms. (b) The student's total. (c) Why they differ.

Solution (a): The first six terms are $3, 7, 11, 15, 19, 23$. Pairing first with last, $\frac{6}{2}(3+23) = 3(26)$. $\boxed{\text{sum} = 78}$

Solution (b): Starting at $n = 0$ adds one more value, $3 + 4(-1) = -1$, on top of the six: $78 + (-1)$. $\boxed{\text{total} = 77}$

(c) — instructor-judged. A correct answer says starting the index at zero adds one extra term — the value the rule would give one step before the sequence begins — so the total shifts by that value; here the phantom term is -1, which is why the wrong total is one smaller. Full credit for naming the extra term and its effect on the total.

---

7. Auditorium: 14 seats in row 1, 2 more per row; the student wrote $14 + 20(2)$. (a) Seats in row 20. (b) Which row has 46?

Solution (a): Row 20 is nineteen rows past row 1: $14 + 19(2) = 14 + 38$. $\boxed{\text{seats} = 52}$

Solution (b): Solve $14 + 2(n-1) = 46$: then $2(n-1) = 32$, so $n - 1 = 16$. Check: $14 + 2(16) = 46$ ✓ $\boxed{\text{row} = 17}$

---

8. $a_1 = 2$, $a_n = a_{n-1} + 7$; the student claimed $a_{100} = 702$ and 9 terms from a_1 to a_{10}. (a) Correct a_{100}. (b) Correct count. (c) What the two share.

Solution (a): Reaching term 100 takes ninety-nine steps: $2 + 99(7) = 2 + 693$. $\boxed{a_{100} = 695}$

Solution (b): From a_1 to a_{10} inclusive: $10 - 1 + 1$. $\boxed{\text{count} = 10}$

(c) — instructor-judged. A correct sentence says both mistakes come from confusing a count of items with a count of gaps between them: the term number counts terms, while the number of steps added counts gaps, and a run of items always has one more item than gap. Full credit for stating the shared cause rather than describing the two errors separately.